

Graduate Algorithms

Quiz 4

September 18, 2026

The bravest thing that you can do in life is ask for help. And even in times when you feel like you're most alone, there's always gonna be someone that you can turn to and ask for help and lean on them. We are a product of the people we surround ourselves with, and friendship is the most important part of life.

Tom Holland

Instructions

- Total marks: 20, Time: 30 minutes.
- There are 2 problems; all are compulsory.
- Write your answers in the answer sheets provided.

Question 1: Discount Buying

10 marks

You have n products in your wishlist with prices: $p_1, p_2, \dots, p_n$.

You can either buy a product individually and pay its full price or use a discount voucher to buy it as part of a group.

You have m vouchers with values: $v_1, v_2, \dots, v_m$.

A voucher with value x allows you to select exactly x products. For those x products, you pay for the $x - 1$ most expensive products and get the cheapest product **FREE**.

Each product can be part of at most one discount group and each voucher can be used at most once.

Professor Caesar claims he has found the optimal strategy, described below.

Algorithm 1 Caesar's Discount

```
1: function CAESARDISCOUNT( $p_1, \dots, p_n, v_1, \dots, v_m$ )
2:   Sort products in decreasing order of price.
3:   Sort vouchers in decreasing order of value.
4:   total  $\leftarrow 0$ 
5:   for each voucher  $v$  in decreasing order do
6:     Assign  $v$  to the  $v$  most expensive remaining products.
7:     Add the sum of the  $v - 1$  most expensive products to total.
8:     Mark all  $v$  products as assigned.
9:   end for
10:  Add prices of all remaining products to total.
11:  return total
12: end function
```

Show or refute: Professor Caesar's algorithm always computes the minimum total cost to buy all products. If the algorithm is incorrect, give a counterexample and describe a fix; if it is correct, argue briefly why (a formal proof is not required).

Question 2: Caesar MST

10 marks

Professor Caesar has designed a new algorithm for computing minimum spanning trees.

For a graph $G = (V, E)$, the set of vertices V is divided into two sets V_1, V_2 such that $V_1 \cup V_2 = V$ and $||V_1| - |V_2|| \leq 1$.

Let E_1 be the set of edges that are incident only to vertices in V_1 and E_2 be the set of edges that are incident only to vertices in V_2 .

The problem is solved recursively on the two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$.

Afterwards, Caesar selects the edge e with minimum weight that crosses the cut (V_1, V_2) in order to connect the two minimum spanning trees into a new spanning tree.

Show or refute: The algorithm computes a minimum spanning tree for G . If the algorithm is incorrect, give a counterexample and describe a fix; if it is correct, argue briefly why (a formal proof is not required).